#3 Quiz Chapter 06, 07, 08&09

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Part A multiple choice (select the one that is best in each case. 1point/question)

1-5 CDBED
6-10 BBBAB
11-15 DDDDA
16-20 DDDCD
21-25 ADCEA
26-30 BECAE
31-35 EECBA
36-37 DD

Part B Short Answer (Write legibly and show all work for all steps in the problem)

38. (5 points)Write the condensed electron configurations and indicate the number of unpaired electrons for the following atoms in their ground states: **(a)** Si (element 14), **(b)** Cr (element 24), **(c)** Hg (element 80) (**d**) Pb^{2+} (**e**) Cu^{+}

Answer:

	Condensed electron configurations	**Number of unpaired electrons**
(a) Si	$[Ne]3s^2p^2$	2
(b) Cr	$[Ar]4s^13d^5$	6
(c) Hg	$[Xe]6s^24f^{14}5d^{10}$	0
(d) Pb^{2+}	$[Xe]4f^{14}5d^{10}6s^2$	0
(e) Cu^{+}	$[Ar]3d^{10}$	0

39. (2 points)What is the wavelength of an electron moving with a speed of 5.97×10^6 $m \cdot s^{-1}$? The mass of the electron is 9.11×10^{-31} kg.

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SOLUTION

Analyze We are given the mass, m, and velocity, v, of the electron, and we must calculate its de Broglie wavelength, λ.

Plan The wavelength of a moving particle is given by Equation 6.8, so λ is calculated by inserting the known quantities h, m, and v. In doing so, however, we must pay attention to units.

Solve Using the value of Planck constant: $h = 6.626 \times 10^{-34}$ J-s

we have the following:

$$\lambda = \frac{h}{mv}$$

$$= \frac{(6.626 \times 10^{-34}\text{ J-s})}{(9.11 \times 10^{-31}\text{ kg})(5.97 \times 10^{6}\text{ m/s})}\left(\frac{1\text{ kg-m}^2/\text{s}^2}{1\text{ J}}\right)$$

$$= 1.22 \times 10^{-10}\text{ m} = 0.122\text{ nm} = 1.22\text{ Å}$$

40. (2 points)What is the wavelength of a golf ball with a mass of 45.9 g moving at a speed of 50.0 m/s?

$$p = mv = 2.295\ kg \cdot m \cdot s^{-1}$$

$$\lambda = \frac{h}{p} = 2.89 \times 10^{-34}\ m$$

41. (2 points)For a hydrogen atom, calculate the energy of the photon absorbed during a transition in which the electron moves from 1*s* to 3*p* orbital.

$$n_f = 3, n_i = 1$$

$$\Delta E = -hcR_H\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) = 1.937 \times 10^{-18}\ J$$

42. (4 points)Write the electron configurations for the following ions, and determine which have noble-gas configurations: **(a)** Co^{2+}, **(b)** Zr^{4+}, **(c)** Ag^{+}, **(d)** S^{2-}

Answer:

(a) Co^{2+},[Ar]3d^7

(b) Zr^{4+},[Kr], noble-gas configuration

(c) Ag^{+}, [Kr]4d^{10}

(d) S^{2-}, [Ar] OR [Ne]3$s^2$3p^6, noble-gas configuration

43. (3 points)Little is known about the properties of astatine, At, because of its rarity and high radioactivity. Nevertheless, it is possible for us to make many predictions about its properties. (**a**) Do you expect the element to be a gas, liquid, or solid at room temperature? Explain. (**b**) Would you expect At to be a metal, nonmetal, or metalloid? Explain. (**c**) What is the chemical formula of the compound it forms with Na?

Answer:

(**a**) solid

The melting points of the halogens increase going down the column.

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(b) Like the other halogens, we expect it to be a nonmetal. According to Figure 7.13, there are no metalloids in row 6 of the periodic table, and At is a nonmetal.

(c) NaAt

44. (3 points) **(a)** What does the term diamagnetism mean? **(b)** Draw the molecular orbital electron configuration for: B_2^+,Ne_2^{2+}, O_2^-. Would you expect them to be diamagnetic or paramagnetic? **(c)** Indicate the bond order change by adding one electron to the ion of the species in part (b).

(a) Substances with no unpaired electrons are weakly repelled by a magnetic field. This property is called diamagnetism.

(b)

B_2^+	$\sigma_{2s}^2\,\sigma_{2s}^{*2}\,\pi_{2p}^1$	paramagnetic
Ne_2^{2+}	$\sigma_{2s}^2\,\sigma_{2s}^{*2}\,\sigma_{2p}^2\,\pi_{2p}^4\,\pi_{2p}^{*4}$	diamagnetic
O_2^-	$\sigma_{2s}^2\,\sigma_{2s}^{*2}\,\sigma_{2p}^2\,\pi_{2p}^4\,\pi_{2p}^{*3}$	paramagnetic

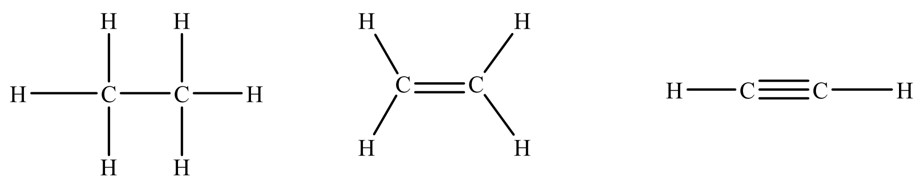
(c)

B_2^+	$0.5 \rightarrow 1$
Ne_2^{2+}	$1 \rightarrow 0.5$
O_2^-	$1.5 \rightarrow 1$

45. (8 points) (**a**) Draw Lewis structures for ethane (C_2H_6), ethylene (C_2H_4), and acetylene (C_2H_2). (**b**) What is the hybridization of the carbon atoms in each molecule? (**c**) Predict which molecules, if any, are planar. (**d**) How many σ and π bonds are there in each molecule?

Answer:

(a)



(b)	sp^3	sp^2	sp
(c)	nonplanar	planar	planar
(d)	$7\sigma,0\pi$	$5\sigma,1\pi$	$3\sigma,2\pi$

46. (3 points)Draw the dominant Lewis structures for each of the following ions or molecules (indicate all resonance structures if there are): **(a)** BrO_3^- **(b)** PH_3 **(c)** SO_3^{2-}

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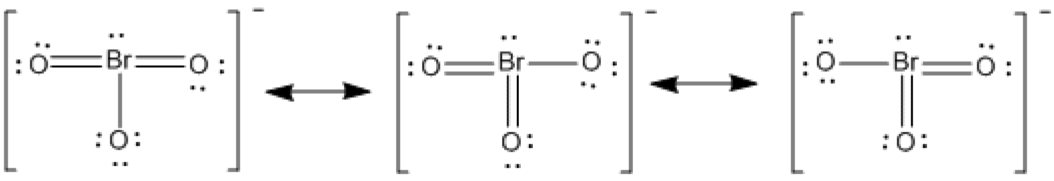
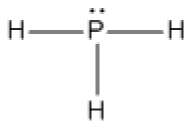
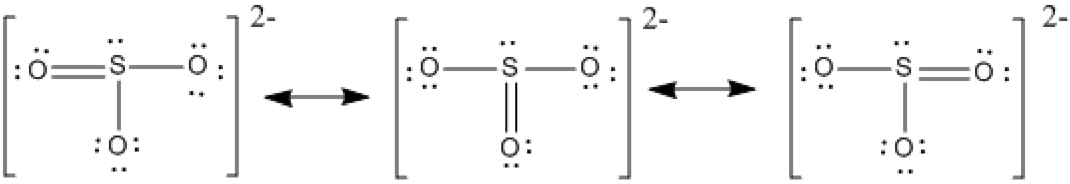
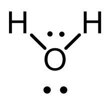
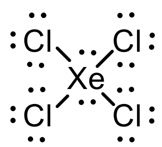
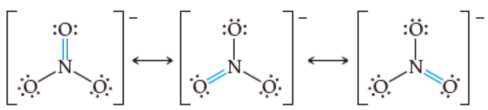
Answer:

(a) 

(b) 

(c) 

47. (6 points)Draw the <u>dominant Lewis structure(s)</u> for each of the following molecules and ions (indicate all resonance structures if there are). Give the electron domain geometry and molecular geometry for each case.
(a) H_2O (b) $XeCl_4$ (c) NO_3^-

Species	Lewis structure	Electron domain geometry	Molecular geometry
(a) H_2O		tetrahedral	bent
(b) $XeCl_4$		octahedral	square planar
(c) NO_3^-		trigonal planar	trigonal planar

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48. (5 points) (a) Draw the molecular orbital of N_2.
(b) The figure that follows shows a sketch of two of the MOs for N_2. What is the label for this MO? How many electrons occupy the MO in the figure?

(c) Is N_2 expected to be diamagnetic or paramagnetic and why?
(d) If one electron is excited from the highest occupied molecular orbital of N_2 to the lowest unoccupied molecular orbital, what is the change of bond order and magnetic behavior?

Answer:

(a)

	B_2	C_2	N_2
σ^*_{2p}			
π^*_{2p}			
σ_{2p}			↑↓
π_{2p}	↑ ↑	↑↓ ↑↓	↑↓ ↑↓
σ^*_{2s}	↑↓	↑↓	↑↓
σ_{2s}	↑↓	↑↓	↑↓

(b)

σ_{2p} and 2 electrons

π_{2p} and 2 electrons

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(c)N_2 have no unpaired electrons and is diamagnetic.

(d)After excitation, the bond order changes from 3 to 2 and the magnetic behavior changes from diamagnetic to paramagnetic.